

CHAPTER

2

Dyeing

5UP8S1

LEARNING OBJECTIVES

- **To understand the classification of dyes**
- **To get an insight into the applicability of individual dyes**
- **To learn about the different stages of dyeing**

2.1 Introduction

Man has known colour since his creation. Nature has been like **Art** with an inexhaustible colour palette with beautiful compositions that inspire man to capture colour on cloths that adorn his body. Primitive man coloured the caves and smeared his body with pigments that he discovered. Ancient people extracted dye from roots, stems, leaves, fruits, berries, flowers, nuts, vegetables, minerals, animals and insects.

Colour plays a vital role in **Human** life. Nature's colour changes with changing season. The colourful wall paintings of Ajanta during the first century AD (CE) and description of natural dyes in *Atharvaveda* provides evidence for the use of dyes in ancient period. People liked to colour everything they use, especially clothes. So they extracted colour from different natural sources such as plants, animals, insects and minerals and applied it on to fabric. The quest for new colours led to the discovery of synthetic dyes which later on became a full-fledged industry.

2.2 Classification of Dyes

Dyes are organic compounds with two components namely Chromophore, which imparts colour and Auxochrome that help in substantivity of dyes. They are classified into natural dyes and synthetic dyes.

2.2.1 Natural Dyes

Natural dyes are colour substances obtained from natural sources. Natural dyes are used for all types of textile dyeing and printing until the middle of nineteenth century. The use of natural dyes were reduced due to the advent of synthetic dyes, though they were economical and posses excellent fastness properties. However, the growing consumer awareness on the harmful impact

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